THE CURE CENTER
FOR ULTRASOUND RESEARCH & EDUCATION

ULTRASOUND PHYSICS AND INSTRUMENTATION SYLLABUS

COURSE CODE: DMS 100

ULTRANSOUND PHYSICS AND INSTRUMENTATION

Instructor:
Telephone:
E-Mail:
Office Hours:
Mon, Wed, Fri: 12pm to 1pm & 5pm to 6pm
Tues: 9am – 5pm
Thurs: 11am – 5pm

Start Date:

Completion Date:

Days and Times:
Monday, Wednesday and Friday

Course Hours:
Morning Session:
8:00 am -10:00am &
10:10am – 12:00 pm
Afternoon Sessions:
1:00pm – 3:00pm &
3:10 pm – 5:00 pm

Class Location:
Classroom #:
Lab#:

Prerequisites:
COM 100 ENGLISH COMPOSITION
MAT 100 GENERAL MATH
PAT 100 PATIENT CARE AND PROFESSIONALISM
LAW 100 MEDICAL LAW & ETHICS

Course Contact Hours: 120
Course length : 15 days
Lecture: 96 hours
Laboratory: 24
Course Delivery: Residential
Pre-requisite: None

Americans with Disabilities Act Guidelines:

Requests for accommodations should be submitted to the campus president. Faculty are not authorized to grant accommodations.

COURSE DESCRIPTION

This is an introductory level course directed towards developing the principles, techniques, knowledge and skills of each student necessary to function as a qualified diagnostic medical sonographer.

This course will introduce students to the fundamental knowledge of acoustic physics, ultrasound principles, and instrumentation.

The students will learn how diagnostic ultrasound works and optimize image acquisition. In addition, they will learn to recognize acoustic artifacts, understand acoustic energy and bio effects while applying the ALARA principle.

Students will apply basic concepts of acoustic physics and sound production, sound propagation, interactions of sound with different media, sound parameters and variables.
Students will learn transducer composition, display modes, imaging artifacts, doppler effect, and hemodynamics. Quality assurance, quality control and the different methods of recording ultrasound will be explored.

Required Student Resources

Textbook:
Title: Understanding Ultrasound Physics
Author(s): Sidney K. Edlman
Publisher: E.S.P. Ultrasound - 2012
ISBN: 0-9626444-5-5

Equipment/Technology

- LCD Projector
- Laptop Computer
- Mindray Ultrasound Machine

Material

- Electronic and Printed Worksheets
- Textbooks
- Visual Aids: DVD Videos, Case Study Reviews, Demonstrations and Hands on Clinical Laboratory Practice.
- Notebook.

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Grade Point Value

Grade	Percentage	Quality Points
A	95 to 100	4.0
A-	90 to 94	3.7
B+	87 to 89	3.3
B	83 to 86	3.0
B-	80 to 82	2.7
C+	78 to 79	2.3
C	73 to 77	2.0
C-	70 to 72	1.7
D+	67 to 69	1.3
D	60 to 66	1.0
F	59 or below	0.0

Methods of Evaluation

Exams	45%
Quizzes	25%
Attendance	10%
Presentation	10%
Homework Assignments	10%
	100%

Teaching Strategies

Lecture
Discussion/Q & A's
Vocabulary and Spelling
Virtual Laboratory Multimedia

Examination Standards

Students are expected to be honest in all academic work. Cheating of any kind is not tolerated. Violations of this principle will result in a grade of zero in the course and additional disciplinary action including expulsion as determined by CURE.

COURSE OBJECTIVES

Upon the successful completion of this courses the student should be able to:

- Perform basic mathematical formula related to ultrasound physics.
- Understand, define abbreviations and terminology as it pertains in ultrasound Physic.
- Describe what is sound, sound beams and Transducer.
- Understand and describe pulsed waves and continuous waves.
- Describe the interaction of sound and different media.
- Understand range equation and dynamic range.
- Understand and describe different frequency used in medical imaging sonography.
- Understand axial and lateral resolution, display modes and 2D imaging.
- Describe beam intensity and describe potential bioeffect.
- Describe ALARA's Policy and Ultrasound equipment safety and quality assurance,
- Describe different transducer and their indication.
- Understand transducer manipulation for best image quality production.
- Describe real time imaging, displays and images processing including harmonics and contrast agents.
- Discuss the processes involved for image display and storage.
- Understand hemodynamics and Doppler effect.
- Describe and understand Doppler effect and Doppler evaluation.
- Identify acoustic artifacts on BW, Spectrum and Doppler images.
- Pass the didactic course with a minimum passing grade of 73.

HOMEWORK

An essential part of every program of study, homework enables students to master course learning objectives and leads toward the achievement of the overall program objectives. Homework refers to self-directed learning activities such as reading assignments, research activities and projects, quiz/exam preparation, knowledge checks and self-assessments, etc.

Late Homework/Make-Up Policy

Late homework will be accepted under the following guidelines:

1. It must be turned in within two business days (Monday through Thursday) after the due date.
2. However, homework turned in after the due date will be penalized 10% for each day it is late up to two calendar days after the due date.
3. Homework submitted after three business days after the due date will earn a zero.

Exams and quizzes - Make up of a quiz or an exam will be allowed under the following extenuating circumstances only:

- Documented illness
- Military duty
- Death of a family member
- Court appearance or jury duty

Documentation of the above circumstances must be provided to the Program Director. ***The ability to make-up a quiz or exam must be approved by the Program Director.***

All assignments should be submitted directly to the instructor.

Classroom Policy

Our goal is to build a respectful learning and work environment that allows for positive communication and teamwork. To support your development, all students must abide by academic policies related to attendance, behavior and professional conduct as published in the current institutional catalog. Special attention is granted to the following items:

- There is to be no eating or drinking in any lab/computer class.
- There is to be no eating in ANY class.
- Cellular phones may only be used during class if/when advised by the instructor and required for a learning activity.
- All students must be in full uniform which includes white or black shoes at **ALL TIMES.**
- **There are no rescheduled exams *unless specified by the professor* with advance notification.**
- The determination to accept late homework assignments is made only by the instructor and occurs only in the event of extraordinary circumstances. If the professor does allow for an extension on assignments, there will be an automatic daily reduction by 10 points, each day beyond the original due date.
- All classes are expected to start and end at a predetermined start and end time.

Accommodations for Students with Disabilities

If you have a special learning need, please contact the Program Director. The Program Director and the Campus President will work with the student to identify reasonable accommodations or adjustments necessary to enable him or her to fully participate in the educational processes.

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Professional Responsibility

Students are expected to aspire to the same high level of ethics and professionalism that is demanded of registered Ultrasound students. This requires students to conduct themselves as professionals and to fully engage in the learning process by preparing for and participating in all classroom, laboratory and clinical activities.

Attendance Policy

- Routine, regular class attendance is a requirement for this course.
- Promptness and dependability are the soft skills that garner special attention

Attendance Rubric:

- Up to 25% will be deducted for each hour missed or for any part thereof (5% for each 10 minutes missed).
- Students are required to meet their professor to discuss matters of tardiness or absence.
- Students are required to notify their instructor via E mail or voicemail to communicate an absence or lateness from class

Attendance is taken each hour of class by the instructor and turned over to the Registrar before the end of the class day. Early departures, tardiest, and class cuts will be recorded in quarter-hour increments and a proportional grade reduction will also be applied. A period of less than 15 minutes will be counted as a quarter-hour of absence. Attendance records are maintained by the Registrar as part of the student's permanent academic record.

- Understand and apply physical principles that apply to obtaining optimal image quality in 2-D real time, Doppler (both spectral and color flow) and static images.

- Understand and apply principles of ultrasound equipment care and maintenance at all locations including CURE facilities, clinical settings, as well as in compromising circumstances such as the emergency room, intensive care units, operating rooms, newborn and neonatal intensive care nursery, burn units, infection control and isolation environments.

Lab Objectives

- Be able to effectively interact with students, instructors, patients, physicians and all other health care professionals of all ages, cultural backgrounds and levels of emotional, mental and physical health and other differing factors without prejudice while learning and performing diagnostic imaging techniques.

- Be able to explain the ultrasound procedure and prepare the patient physically, mentally and emotionally for the sonographic exam as well as provide patient education on the diagnostic use of medical ultrasound specific to their exam prior to the start of the exam.

- Be able to elicit cooperation from the patient to enhance imaging quality.

- Provide any safety and comfort measures for the patient during the exam.

- Be able to effectively apply aseptic, bacteriostatic and sterile techniques in the care of the patient, themselves, others and the ultrasound equipment.

- Understand and apply appropriate principles of body mechanics, ergonomics and safety measures in the care and prevention of injury to the patient, themselves and others.

- Be familiar with and be able to work around all medical equipment which may accompany or be in use for a patient and their role concerning that equipment which will include, but not be limited to, intravenous lines, oxygen tubes and tanks, Foley catheters, chest tubes, feeding tubes, telemetry, ventilators, etc.

- Be able to recognize the signs and symptoms of a medical emergency, respond and act accordingly on the patient's behalf and assume their role in that situation based on the law and specific protocol of that medical institution.

- Understand and apply all physical and instrumentation principals as it applies to obtaining optimal images, which will include, but not be limited to, biological effects, body habitus, image artifacts, quality control, system gain adjustments, focal zone, depth, transducer selection, as well as Doppler principals and application.

- Understand the importance of and be able to obtain necessary and pertinent information from the patient's chart in order elicit to know the reason for the exam and to determine what the ordering physician is "ruling out", as well as to provide the area of concentration the sonographer is to focus on.

- Be able to interpret all correlating diagnostic results in order to obtain an optimal exam and to provide a clear and thorough preliminary interpretation for the physician.

- Be able to recognize, identify, distinguish, obtain and label all normal and abnormal anatomy, associated vasculature, ligamentous structures, benign and pathologic entities and related structures according to designated ultrasound protocol.

- Recognize, distinguish and interpret normal and abnormal sonographic patterns, related clinical signs and symptoms, normal anatomic variants, benign entities and pathology in obtaining PHYSICS images according to designated ultrasound protocol.

- Perform and record normal, abnormal and pathologic data in the form of routine optimal images, appropriate measurements and applied spectral and color Doppler analysis according to designated protocol suitable for interpretation by a reading physician to formulate a diagnosis.

- Understand and assess the principles associated with suboptimal imaging, evaluate under what circumstances these principles and results occur and determine the methods which are best to remedy these pitfalls.

- Prepare clear, precise, thorough and objective preliminary interpretations.

Teaching Strategies

- The instructor should emphasize the importance of the clinical problem and the role of sonography in detection of pathology.

- The instructor should stress the importance of the scope of knowledge required and how that knowledge relates to skills of the sonographer in evaluation of the patient.

- In obtaining the patient's clinical history, the instructor should provide examples of the type of information found in these sections for pathology and stress the importance of confidentiality of this information.

- The instructor should present an explanation of all the directional planes and the anatomy in the longitudinal/sagittal plane initially. This should be followed by the coronal plane, since this is typically the

presentation in general anatomy. The educator should relate the r structures to the coronal, longitudinal and transverse planes and stress the importance of using surrounding anatomical landmarks to properly identify all anatomical structures.

- The physiologic basis for clinical laboratory tests should be emphasized.

- The students must realize the importance of examining the results of related imaging studies before the performance of the ultrasound exam. The advantages and disadvantages of each procedure should be discussed along with the importance of reviewing the reports and films before performing the ultrasound examination. Students without an imaging background may need additionally information to improve comprehension.

- Scanning protocols and techniques are usually presented in a demonstrated lesson in conjunction with laboratory scanning sessions. The instructor should emphasize the importance of all these positioning and technical factors by providing clinically related examples. The importance of obtaining serial images should be stressed in terms of encouraging students not to miss pathology and reinforcing the knowledge of anatomy covered in previous lectures. Different scanning protocols in specific departments should also be covered; the student must adopt a standard basic protocol based on national guidelines, which then can be modified for use in a specific department. The instructor should demonstrate all skills required for the performance of the examination before student practice sessions.

- In teaching the normal sonographic patterns involved in sonography, the instructor should stress the importance of verifying that other organs have a normal pattern in comparison with abnormal patterns and fluid collections. It should also be stressed that the patient is an important factor in relating the area or quadrant of pain before the exam.

- In discussing congenital anomalies and anatomic variants, the educator should help the student recognize body habitus when determining the best acoustic window to use for examination.

- Students often have difficulty understanding what is meant by an “equivocal measurement and that the normal range varies within the age of the patient. The instructor must explain all concepts and stress that sonography research affects normal size ranges, normal sonographic pattern classification, and abnormal patterns associated with specific pathologic conditions. Normal measurement ranges should be decided by the program and be consistent within all handout material and testing.

- In learning pathologic processes, the student must initially comprehend and remember the specific information for each type of pathology. Therefore, the instructor should assist the students in attaining the knowledge at a cognitive level by giving clinically related examples. Since this requires a lot of memorization on the part of the student, the use of short oral or written quizzes may prove beneficial. The instructor can begin the process for forming differential diagnosis but concentrate on the most typical pattern and symptoms.

- The instructor must emphasize the importance of ultrasound physics and instrumentation in the clinical setting when discussing technical pitfalls. This relationship will assist the student in comprehending physics by providing concrete, clinical examples related to the abstract theory. The instructor should also stress how many of these artifacts and pitfalls can degrade the diagnostic quality of the examination and/or lead to a misdiagnosis.

- The instructor should help the student understand the guidelines, which they must follow in writing preliminary interpretations of an exam & how to stay within those guidelines while practicing as a sonographer and not a legal diagnostician.

Teaching Supplements

- Anatomical charts, illustrations, anatomical models, DVDs on anatomy and diagrams in coronal, longitudinal and cross-sectional anatomy.

- Interesting case studies and/or examples to get-the-point-across, make the information more meaningful, interesting and for complete comprehension.

- Blank copies of physician's order sheets, patient's history section (H&P), vital signs graph/chart, laboratory results and nurses notes can be used as an overhead transparency, in a slide or PowerPoint presentation in order to better understand and become familiar with the patient's chart and the process of obtaining the patient's clinical history/picture. Students obtaining the H&P by role-playing can also be supervised by the educator.

- For better understanding of relational anatomy, correlate the diagrams within the teaching text, obtain slides or PowerPoint presentation of cadaver sections or CT/MRI images and sonograms illustrating anatomy in all planes of section. Students may also be divided into small groups and asked to reconstruct all pertinent anatomy with modeling clay and pipe cleaners. When finished, the instructor can then cut the models in coronal, transverse and longitudinal sections and review the displayed anatomy.

- Cases which include examples of related imaging procedures and the sonographic examination will provide excellent reinforcement for learning "related imaging procedures" and how they help to formulate to patient's clinical picture.

- The instructor can use slides, videos, DVDs, PowerPoint presentations and sonographic case studies that illustrate the various types of pathology. A clinical history can be provided to the student along with a sonographic case for students to identify the pathology. A game of "Pictionary" can also be used where students can try to draw the sonographic feature of that pathology for others to identify. Try writing different pathologies on pieces of paper and place in a bowl or bag for students to draw from. They then have to describe the clinical signs and symptoms of that pathology for the class to identify and come up with the answer. This method teaches both the student quizzing and the students coming up with the answer. The instructor can have student's role play different pathologies as they role-play the patient with signs and symptoms for the sonographer. Have the students look up and obtain photos of different pathologies to present to the class themselves. The instructor should emphasize the importance of "visual memory" or "mental rehearsal" in the process of memorization and learning sonographic images.

- The instructor should provide various examples of technical pitfalls and artifacts with reference to the text, slides, sonographic images, cases studies, transparencies or PowerPoint presentation. The instructor could provide mystery cases where the students must identify the technical pitfall or artifact.

The course syllabus is a general plan for the course; deviations may be necessary and will be announced.

ULTRASOUND PHYSICS AND INSTRUMENTATION SYLLABUS

COURSE OUTLINE

Below is the course schedule which identifies lecture topics, assignments and other course activities and the homework required for completion each week.

DAYS	Lecture Topic	In-Class Activity/Lab Assignment	Homework/Out of Class
DAY 1	The Basics: Mathematics	Introduction to course, class policies, and Syllabus: **CHAPTER 1: THE BASICS** • Metric and measurements • Fractions, decimals percentages and interconversions • Logarithms and exponents • Algebraic functions	Understanding Ultrasound Physics Textbook Vol. 4, by Sidney K. Edelman Chapter 1 "The Basics" pg. 2-6
DAY 2	Principles of Sound Principles of Waves	**CHAPTER 2: SOUND** • Sound Waves • Acoustic Variables • Acoustic Parameters • In Phase and Out of Phase waves **CHAPTER 3: DESCRIBING SOUND WAVES** • Period • Frequency • Wavelength • Propagation speed **CHAPTER 4: DESCRIBING PULSED WAVES** • Pulse Duration • Spatial Pulse Length • Pulse Repetition Period • Pulse Repetition Frequency • Duty Factor	Understanding Ultrasound Physics Textbook Vol. 4, by Sidney K. Edelman Chapter 2 "Sound" pg. 11-16 Understanding Ultrasound Physics Textbook Vol. 4, by Sidney K. Edelman Chapter 3 "Describing sound waves" pg. 20-41 Understanding Ultrasound Physics Textbook Vol. 4, by Sidney K. Edelman Chapter 4 "Describing Pulsed waves" pg. 49-68
DAY 3	Range Equation and Transducers	**CHAPTER 6: INTERACTION OF SOUND AND MEDIA** • Decibels(db) • Attenuation • Reflection and Transmission • Impedance **CHAPTER 7: RANGE EQUATION** • Time -of-Flight or Go-Return Time **CHAPTER 8: TRANSDUCERS** • Basic Transducer Construction • Transducer Frequencies	Understanding Ultrasound Physics Textbook Vol. 4, by Sidney K. Edelman Chapter 6"Interaction of Sound and Media" pg. 77-106 Understanding Ultrasound Physics Textbook Vol. 4, by Sidney K. Edelman Chapter 7 "Range Equation" pg. 107-111 Understanding Ultrasound Physics Textbook Vol. 4, by Sidney K. Edelman Chapter 8 "Range Equation" pg. 114-126 **QUIZ #1**

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DAYS	Lecture Topic	In-Class Activity/Lab Assignment	Homework/Out of Class
DAY 4	Sound Beams and Resolution QUIZ	**CHAPTER 9: Sound Beams** • Shape of a Sound Beam • Anatomy of a Sound Beam • Determining Focal Length • Sound Beam Divergence • Spherical Waves • Huygens' Principle **CHAPTER 10: AXIAL AND LATERAL RESOLUTION** • Axial Resolution • Lateral Resolution • Focusing	Understanding Ultrasound Physics Textbook Vol. 4, by Sidney K. Edelman Chapter 9 "Sound Beams" pg. 130-144 Understanding Ultrasound Physics Textbook Vol. 4, by Sidney K. Edelman Chapter 10 "Axial and Lateral Resolution" pg. 145-154
DAY 5	Display Modes	**CHAPTER 11: DISPLAY MODES** • A-Mode—Amplitude • B-Mode---Brightness • M-Mode—Motion **CHAPTER 12: TWO-DIMENSIONAL IMAGING** • Mechanical Transducer • Linear Phased Arrays • Annular Phased arrays • Linear Sequential Arrays • Vector Arrays • Slice Thickness or Elevational Resolution	Understanding Ultrasound Physics Textbook Vol. 4, by Sidney K. Edelman Chapter 11 "Display Modes" pg. 159-162 Understanding Ultrasound Physics Textbook Vol. 4, by Sidney K. Edelman Chapter 12 "Two-Dimensional imaging" pg. 167-188
DAY 6	Real-Time Imaging and Instrumentation	**CHAPTER 13: REAL -TIME IMAGING** • Temporal Resolution • Imaging Depth • Number of Pulses per Image **CHAPTER 14: PULSED ECHO INSTRUMENTATION** • Pulser • Beam Former • Receiver • Output Power Vs. Receiver Gain **MIDTERM TEST**	Understanding Ultrasound Physics Textbook Vol. 4, by Sidney K. Edelman Chapter 13 "Real-Time Imaging" pg. 198-201 Understanding Ultrasound Physics Textbook Vol. 4, by Sidney K. Edelman Chapter 14 "Display Imaging and Processing" pg. 239-261 **MIDTERM TEST**
DAY 7	Displays and Image Processing	**CHAPTER 15: DISPLAYS AND IMAGE PROCESSING** • Display Control • Analog and Digital Image Data • Magnification • Coded Excitation • Spatial Compounding • Frequency Compounding • Recording and Archiving Techniques	Understanding Ultrasound Physics Textbook Vol. 4, by Sidney K. Edelman Chapter 15 "Display and Image Processing" pg. 240-261

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DAYS	Lecture Topic	In-Class Activity/Lab Assignment	Homework/Out of Class
DAY 8	Dynamic Range	**CHAPTER 16: DYNAMIC RANGE** • Dynamic Range of System Components • Number of Choices	Understanding Ultrasound Physics Textbook Vol. 4, by Sidney K. Edelman Chapter 16 "Dynamic Range" pg. 265-268
DAY 9	Hemodynamics	**CHAPTER 18: HEMODYNAMICS** • Flow • Laminar Vs Turbulent • Energy Gradient • Forms of Energy	Understanding Ultrasound Physics Textbook Vol. 4, by Sidney K. Edelman Chapter 18 "Hemodynamics" pg. 286-300
DAY 10	Hemodynamics	**CHAPTER 18: HEMODYNAMICS** • Stenosis • Venous Hemodynamics • Hydrostatic Pressure • The Doppler Equation	Understanding Ultrasound Physics Textbook Vol. 4, by Sidney K. Edelman Chapter 18 "Hemodynamics" pg. 286-300 **QUIZ # 2**
DAY 11	Doppler	**CHAPTER 19: DOPPLER** • Speed vs. Velocity • The Doppler Equation • Bidirectional Doppler	Understanding Ultrasound Physics Textbook Vol. 4, by Sidney K. Edelman Chapter 19 "Doppler" pg. 303-340
DAY 12	Doppler	**CHAPTER 19: DOPPLER** • Continuous Wave Doppler • Pulsed Wave Doppler • Color Flow Doppler • Doppler Artifacts • Spectral Analysis •	Understanding Ultrasound Physics Textbook Vol. 4, by Sidney K. Edelman Chapter 19 "Doppler" pg. 303-340
DAY 13	Optimizing doppler imaging	**CHAPTER 20: OPTIMIZING DOPPLER IMAGING** • Normal Incidence • Color Doppler Gain • Spectral Doppler Gain • Aliasing • Wall Filter Vs. Increasing Scale •	Understanding Ultrasound Physics Textbook Vol. 4, by Sidney K. Edelman Chapter 20 "Optimizing Doppler Imaging"
DAY 14	Optimizing doppler imaging and Artifacts	**CHAPTER 21: ARTIFACTS** • Image Characteristics • Comet Tail • Shadow • Mirror Image • Lobes • Refraction	Understanding Ultrasound Physics Textbook Vol. 4, by Sidney K. Edelman Chapter 21 "Artifacts" pg. 355-378
DAY 15	Quality Assurance and Bioeffects	• **CHAPTER 22: QUALITY ASSURANCE** • Major Principles of Medical Ethics • Informed Consent • Patient-Sonographer Interaction **CHAPTER 24: BIOEFFECTS** • Converting Sound energy into heat • Risk-Benefit Relationship • Electric Safety Final Review and Final exam	Understanding Ultrasound Physics Textbook Vol. 4, by Sidney K. Edelman Chapter 22 - "Quality Assurance" pg. 379-390 Understanding Ultrasound Physics Textbook Vol. 4, by Sidney K. Edelman Chapter 24 – "Bioeffects" pg. 401-402